

ENVIRONMENTAL DATA SCIENCE (CS4086)

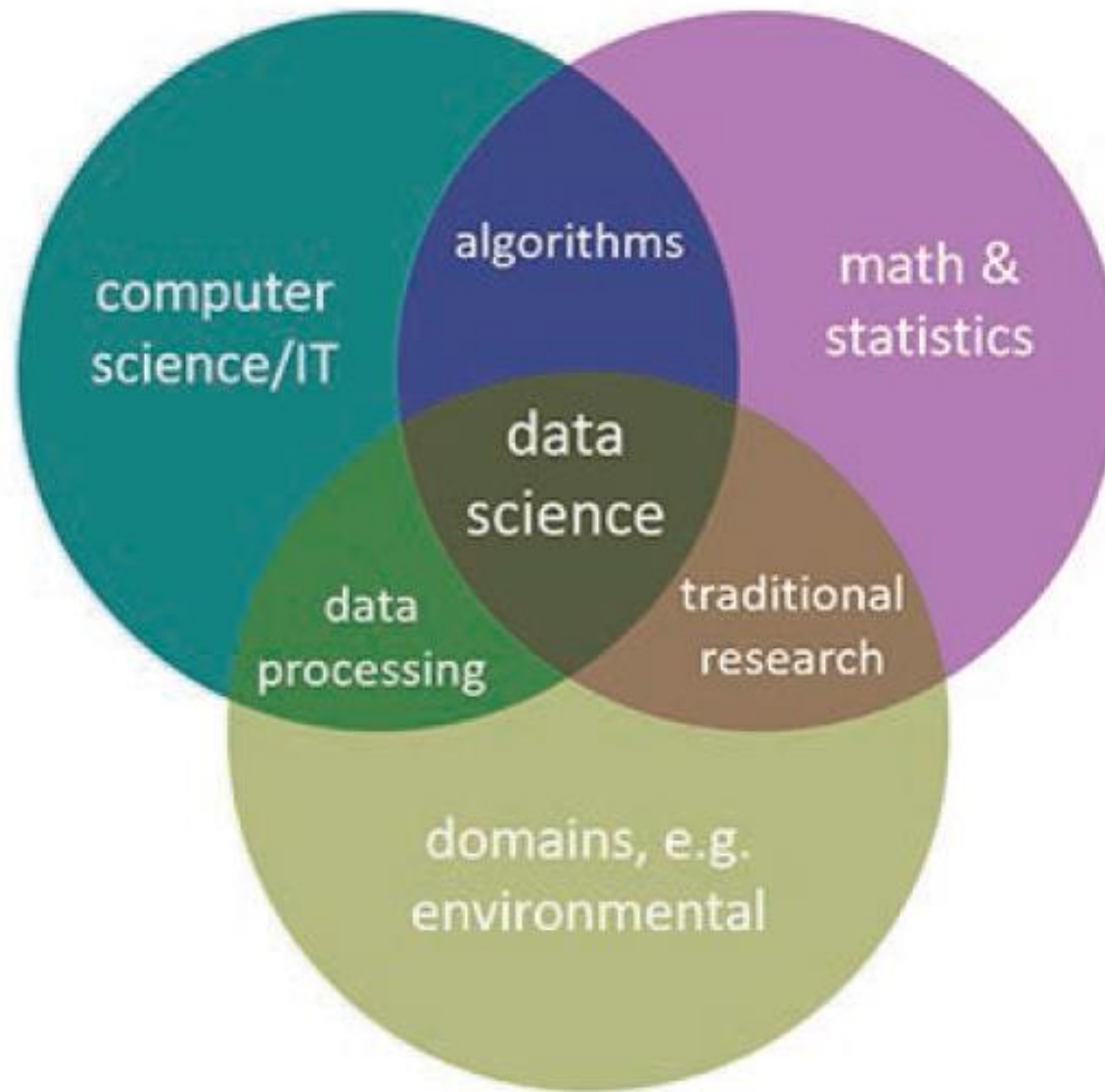
INTRODUCTION & BACKGROUND

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- ▶ Data science is an interdisciplinary subject that draws information and insights from noisy, structured, and unstructured data using scientific methods, processes, algorithms, and systems.
- ▶ Environmental data is a classic example of an application that benefits greatly from a data science approach, as it consists of continually expanding collections of data from automated sensors in space and time domains.
- ▶ Data science applied to environmental science research is known as environmental data science. Data science is often thought of as the nexus of mathematics, statistics, computer science/IT, and certain study fields; in this instance, it's environmental data-science.

INTRODUCTION



Since environmental data can comprise a wide range of information, including environmental measurements in the space and time domains, a variety of approaches can be used in environmental research.

▶ **DATA ANALYSIS & TRANSFORMATION METHODS:**

- ▶ -importing
- ▶ -recognition
- ▶ -filtering
- ▶ Data-joints
- ▶ Reorganizing data

VISUALIZATION

- graphics
- maps
- imagery

SPATIAL ANALYSIS

- vector & raster spatial analysis
- spatial statistics
- image analysis

ENVIRONMENTAL DATA AND METHODS

▶ **STATISTICAL SUMMARIES TESTS & MODELS:**

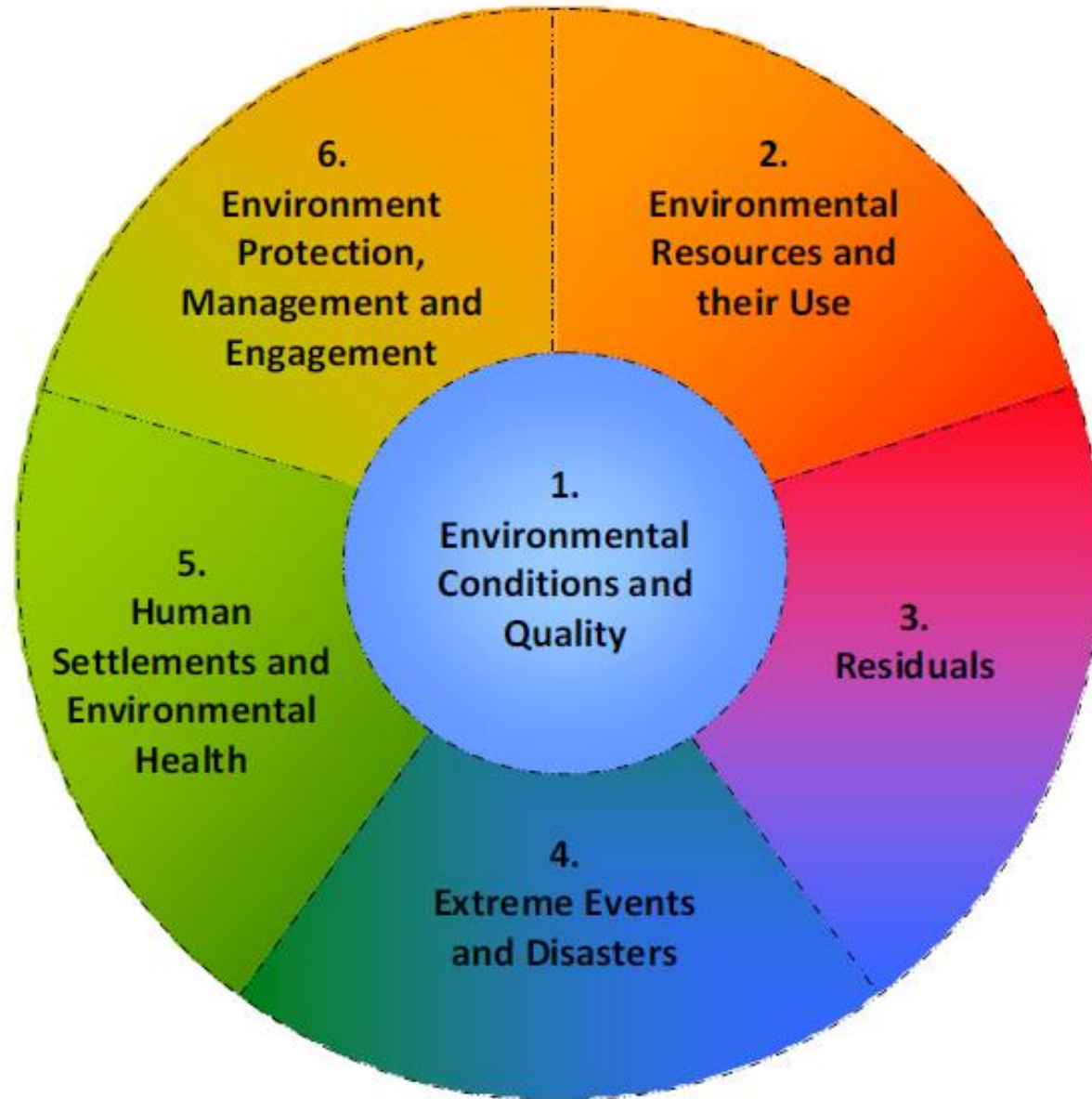
- ▶ -statistical tests
- ▶ -classification models
- ▶ -physical, statistical & machine learning models
- ▶ -artificial intelligence, data mining etc.

▶ **TEMPORAL DATA & TIME SERIES**

- ▶ Analyzing & visualizing long-term environmental data
- ▶ Analyzing & visualizing high-frequency data from loggers

ENVIRONMENTAL DATA AND METHODS

- **Provide information about the environment:**
 - Most important changes over time and across locations
 - Main factors that influence them
 - **Purposes:**
 - Improve knowledge of the environment
 - Support evidence-based policy and decision making
 - Inform the general public and specific user groups
- 



Environmental DATA:

- Unprocessed observations and measurements about the environment and related processes
- Sources: Surveys, Administrative, Research,...

Environment STATISTICS:

- Statistical methods, standards, procedures applied
- Role of environment statistics to process environmental data into meaningful statistics

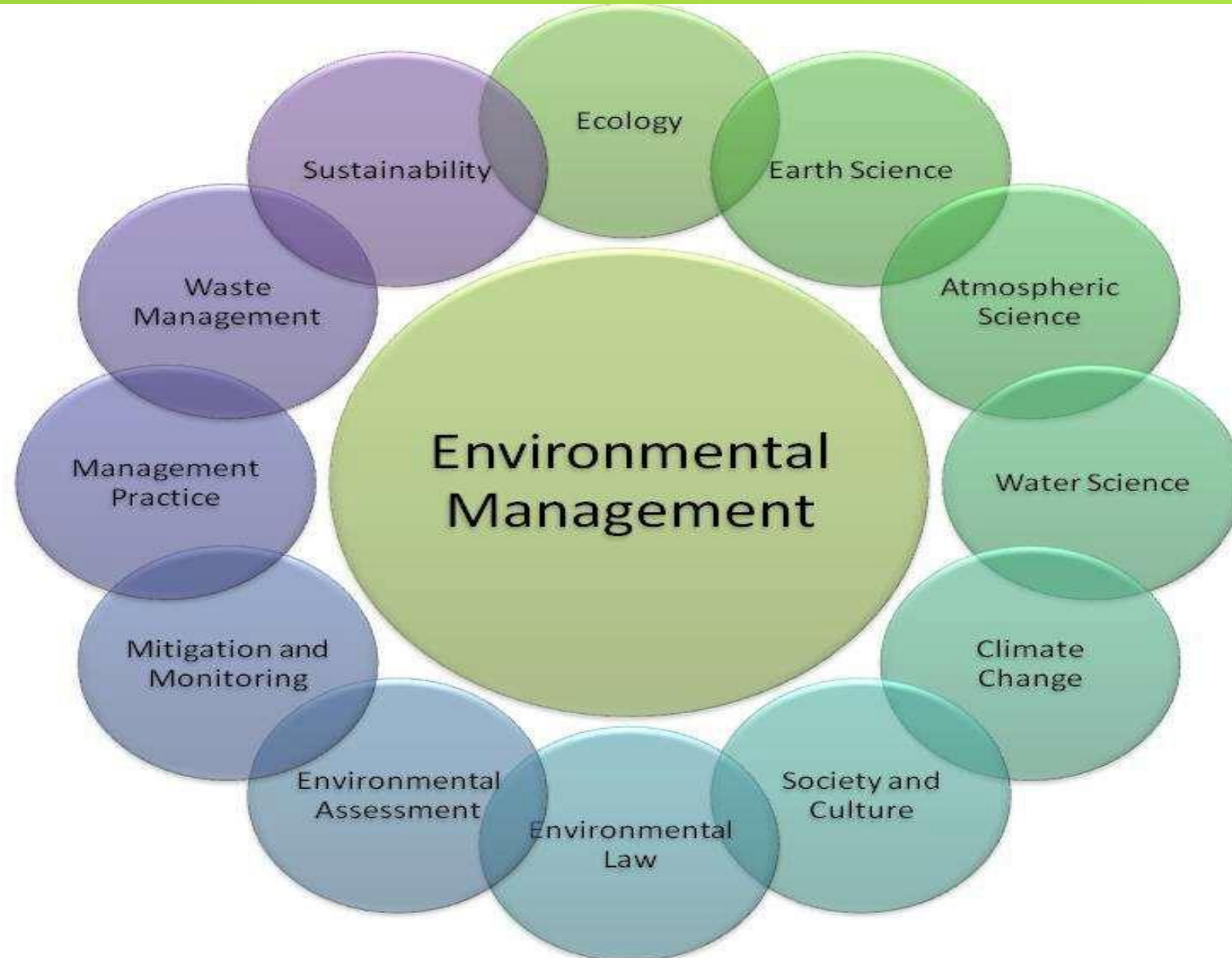
Environmental INDICATORS:

- Synthesize and present complex statistics
- Define objectives, assessing present and future directions etc.
- Frameworks are e.g. DPSIR, MDGs, SDIs, Climate Change

Biophysical aspects of the environment, those **aspects of the human sub-system** that directly influence the state and quality of the environment, and the **impacts of the changing environment** on the human sub-system.

Includes interactions with the environment, and among the environment, human activities, and natural events.







- ▶ ❖ Applied for both short term and long term concerns.
- ▶ ❖ Integration of natural and social sciences.
- ▶ ❖ Integration of policy making and planning.



CHARACTERISTICS OF ENVIRONMENTAL MANAGEMENT

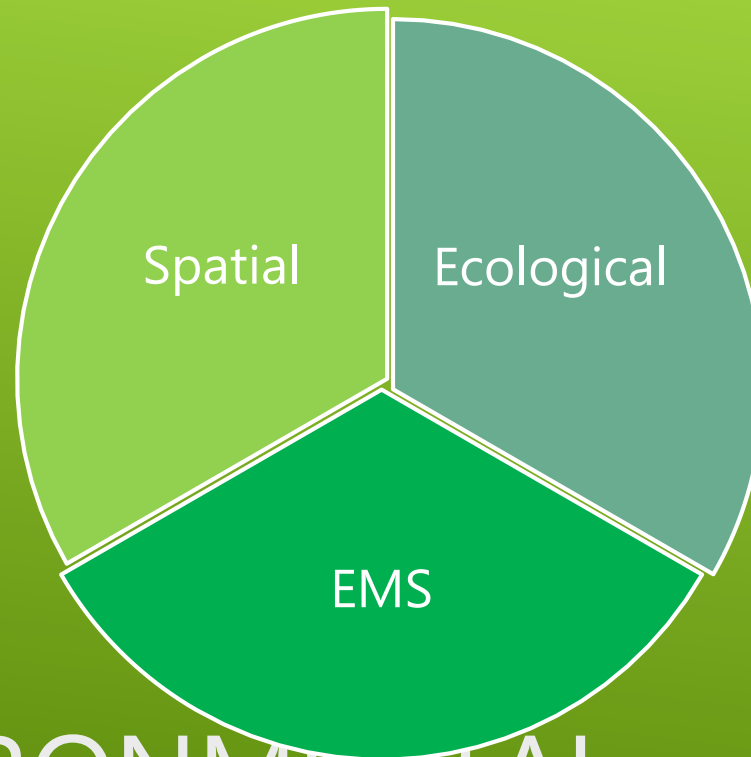
- ❑ To prevent and solve environmental problems.
- ❑ To develop research and monitoring.
- ❑ To warn threats and identify opportunities.
- ❑ To suggest measures to resource
 - ▶ conservation.
- ❑ For long term/short term sustainable developments.
- ❑ Develop a strategy to solve the environmental issues.



SIGNIFICANCE OF ENVIRONMENTAL MANAGEMENT

There are 3 approaches of environmental management-

1. Spatial approach
2. Ecological approach
3. Environmental Management System (EMS) Approach



APPROACHES OF ENVIRONMENTAL MANAGEMENT

Spatial Approach considers the management of environment at 3 levels-

1. Local level
2. Regional level
3. Global level

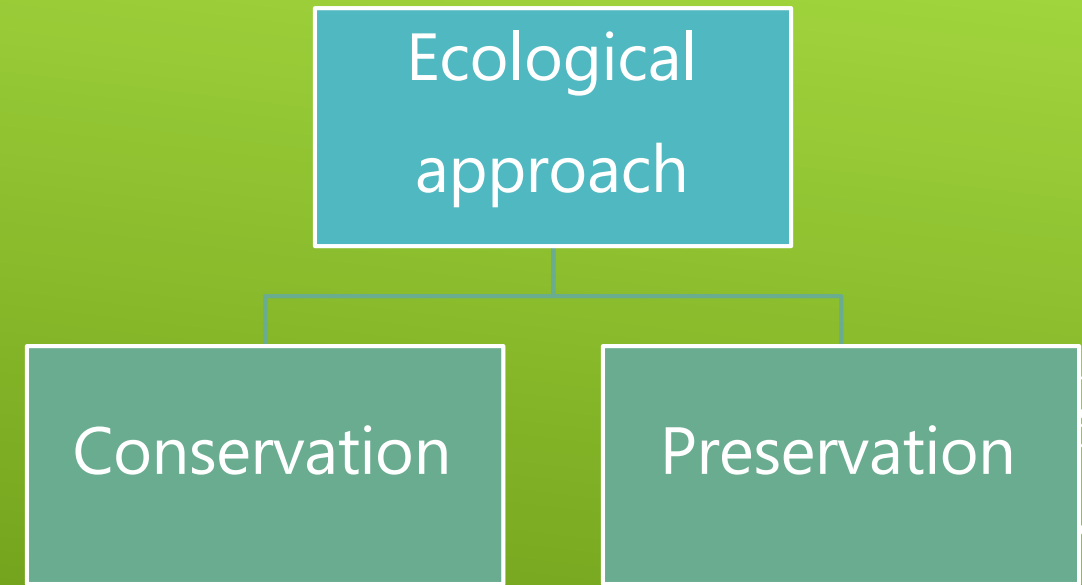
At global level, the spatial dimension and management strategies may be chalked out on the basis of needs, aspirations, perceptions, priorities and inhibitions of the individuals, society and government.

SPATIAL APPROACH



► Ecological approach considers the management of ecological resources to save the environment. It has two methods to manage ecology.

1. Conservation: To utilize the natural environment for social needs but a simultaneous effort to maintain the ecological balance.
2. Preservation: To save some species or natural environments from social utilization in order to protect them.



ECOLOGICAL APPROACH

EMS has been defined by ISO 140001 as-

“The part of the overall management system that includes organizational structure, planning activities, responsibilities, practices, procedures, processes and resources for developing, implementing achieving and reviewing the environmental policy.”



ENVIRONMENTAL MANAGEMENT SYSTEM (EMS) APPROACH

Environmental Planning is a decision-making process that considers the environmental, social, political, economic, and governance factors that can affect development. Its goal is to manage the relationship between natural systems and human systems in the present and the future.



ENVIRONMENTAL PLANNING

- ▶ In general, environmental data are needed to delineate (Harmancioglu *et al.* , 1992;
- ▶ Whitfield, 1988):
- ▶ a) the general nature and trends in characteristics of environmental processes for a
- ▶ better understanding of these processes;
- ▶ b) the effects of natural and man-made factors upon the general trends in
- ▶ environmental processes;
- ▶ c) the effectiveness of pollution control measures;
- ▶ d) the compliance of environmental quality characteristics with established quality
- ▶ standards for eventual purposes of enforcing quality control measures.

NEED OF ENVIRONMENTAL DATA

- ▶ Furthermore, data are the essential inputs for:
- ▶ a) environmental impact assessment;
- ▶ b) assessment of general quality conditions over a wide area or "general surveillance" ;
- ▶ c) modeling of environmental processes.

NEED OF ENVIRONMENTAL DATA

- ▶ Natural disaster monitoring, forecasting, and management approaches have advanced significantly during the last century as a result of technological advancement.
- ▶ These new technologies include the application of artificial intelligence, which, because of its capacity to predict future occurrences, has the potential to make a significant impact in reducing the human and material costs of such disasters.
- ▶ Scientists were able to forecast tsunami amplitudes without defining the triggering earthquake by merging **Global Navigation Satellite System data with AI**, avoiding concerns such as magnitude saturation, which is prevalent in seismic-based detection systems.
- ▶ **An earthquake early warning system** that uses artificial intelligence (AI) to forecast how the ground would move during a tremor and can provide several seconds' notice that the shaking is imminent. On the West Coast of the United States, a comparable system that employs more traditional processing power already exists.
- ▶ **Google Research employees** have created an artificial intelligence algorithm that can anticipate the weather within 5 minutes. While this system, known as "**ML for Precipitation Nowcasting from Radar Images**," is ideally adapted to weather forecasting, it may also be useful in predicting future natural disasters.

ROLE OF DATA SCIENCE IN ANALYZING ENVIRONMENTAL ISSUES-EXAMPLES

- ▶ A group of **Stanford University academics** has created a new machine learning method called Earthquake Transformer with the explicit goal of addressing this challenge. This algorithm is capable of giving a degree of accuracy comparable to human assessments and of identifying a greater number of earthquakes, particularly low-intensity earthquakes that are typically missed by existing detection methods.
- ▶ This improved detection capability was obtained by modifying the top medical image processing and audio capture algorithms to detect even the weakest signals. Recent **disaster management assessments** have focused on AI applications in disaster management as well as the usage and promise of big data in natural catastrophe management.
- ▶ Disaster management AI applications include **hazard risk assessment**, vulnerability assessment, early warning systems, disaster detection, event mapping, damage assessment, disaster rescue and relief, and resource allocation. There is a growing trend in disaster management to use AI to evaluate and analyze massive data from multiple data sources to make educated decisions.

ROLE OF DATA SCIENCE IN ANALYZING ENVIRONMENTAL ISSUES-EXAMPLES

- ▶ **Environmental Data Analysts** enable the worldwide community of environmental researchers to speak the same language and non-scientists to understand the data presented by the community. Environmental Data Analysts serve as both **communicators and educators**.
- ▶ An environmental data analyst should have an understanding of **data management** and the ability to develop spreadsheets and databases utilizing superior computer abilities, detail-oriented and methodical, with exceptional problem-solving abilities.
- ▶ They can collect a large amount of data on a certain environmental issue; both **quantitative and qualitative data** are frequently necessary.
- ▶ They can analyze environmental data and present it to companies, clients, or government agencies, and apply testing and statistical analysis to the environmental data to produce results.
- ▶ They can prepare models based on **artificial intelligence and machine learning** to pre-predict the devastated natural hazards to save populations at earlier stages with **mitigation methods**

ROLE OF AN ENVIRONMENTAL DATA ANALYST

- ▶ **Environmental Intelligence (EI)** is an emerging study subject of strategic worldwide relevance, according to evidence from academics, policymakers, and businesses.
- ▶ Its main goal is to comprehend and respond to the complex relationships that exist between the **environment, climate, natural ecosystems, human social and economic systems, and health**.
- ▶ The following are key areas of concern:
 - ▶ **Changes in the Climate** (e.g., resilient cities, clean power, smart transport options, sustainable production).
 - ▶ **Natural Hazards, Biodiversity and Conservation** (for example, habitat conservation and restoration, pollution control, and developing natural capital) (e.g., predicting and forecasting, early warning systems, resilient infrastructure).
 - ▶ **Health and the Environment** (e.g., air quality, water, sanitization, social and environmental inequalities).

ENVIRONMENTAL INTELLIGENCE (EI)

- ▶ Between 2020 and 2030, the demand for environmental data analysts is predicted to rise by 8% globally.
- ▶ Government, industry, and charity all rely on **statistics** considerably more than they used to, and it may soon become a skills shortage. The majority of this is projected to be in **commercial services**; environmental data analysis is unlikely to be in such great demand on its own.
- ▶ However, primary data analysis may be in high demand, particularly in specialized IT capabilities such as GIS.
- ▶ **Environmental Data Analysts** gather and evaluate information on a variety of environmental concerns.
- ▶ Many **corporations and government** agencies rely on conclusive data once it has been reviewed to offer proof for managerial, financial, and project decisions.

JOB DEMAND

- Research scientist (Environmental Science)
- Environmental Scientist
- Atmospheric Data Scientist
- Environmental Data Scientist
- Environmental Systems Data Scientist
- Environmental Risk-Data Management Analyst
- Environmental Statistician
- Data Scientist (Sustainability)
- Environmental Modeling Scientists
- Environmental Modeler-Machine Learning
- Applied Scientist for Sustainability
- Sustainability Data Analyst
- Environmental Data Analyst

JOB DEMAND